EXP 3: Develop a flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system.

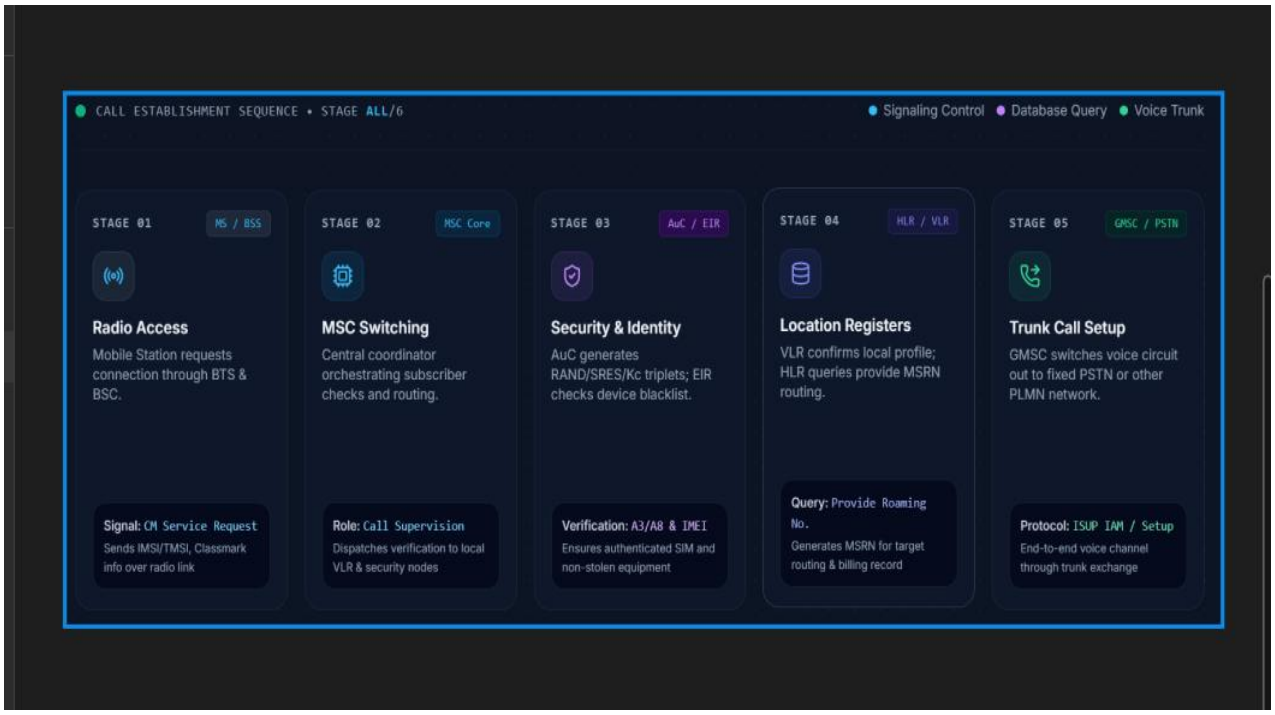
Aim

To design, model, and document the operational call-flow and functional architecture of the Network Switching Subsystem (NSS) in a mobile cellular system using Figma.

Procedure:

1. **Initialize Figma Canvas:** Launch Figma, create a new design file, and insert a Desktop frame (1920 × 1080 px) with a subtle grid layout.
2. **Define Design Tokens:** Establish a color palette separating entities: Blue for MSC/switching, Purple for databases (HLR, VLR, AuC, EIR), Amber for decision nodes, and Green/Red for terminal states.
3. **Build Core Shape Components:** Construct reusable vector master components (Ctrl + Alt + K) for process rectangles, database cylinders, and decision diamonds.
4. **Layout Functional Swimlanes:** Group the workspace into four horizontal lanes using Auto Layout: *User/Radio (MS/BSS)*, *Core Switching (MSC/GMSC)*, *Databases & Security*, and *External Network (PSTN)*.
5. **Place Mobile-Originated Sequence Nodes:** Place the initiating blocks starting from the Mobile Station (MS) signaling the Base Station Subsystem (BSS) and routing into the MSC.
6. **Integrate Database Verification Blocks:** Position parallel lookup blocks connecting the MSC to the VLR for active subscriber records and the HLR for profile verification.
7. **Incorporate Security Logic:** Add decision branches connecting the MSC/VLR to the Authentication Center (AuC) for cipher triplets and the Equipment Identity Register (EIR) for IMEI validation.
8. **Map Routing and External Handover:** Add routing resolution steps directed toward the Gateway MSC (GMSC) and out to the Public Switched Telephone Network (PSTN).
9. **Route Orthogonal Connectors:** Use the Connector Tool (Shift + C) to draw elbowed paths connecting the nodes, snapping directly to anchor points.
10. **Annotate Path Logic & Labels:** Label decision branches with explicit conditional tags ([Authorized / Valid IMEI] and [Authentication Failure / Blacklisted]), and export the final flowchart as a high-resolution PNG/PDF.

Design:



Result

The operation of the Network Switching Subsystem (NSS)—encompassing subscriber authentication, database querying (HLR/VLR/AuC/EIR), and mobile-originated call establishment—was successfully modeled into a clear, standardized, and fully annotated flowchart in Figma.